

A Linear Surrogate-Based Algorithm for Fitting Gaussian Mixture Functions

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Abstract—Gaussian Mixture Function (GMF) is a widely utilized model for analyzing and elucidating experimental data in science and engineering, where the fitting of GMF with noisy observations is usually rendered a complicated nonlinear regression problem due to the underlying linear superposition of Gaussian components. Classical Newton-type solutions rely on derivatives of the regression objective to facilitate convergence, which are general-purpose and can be inefficient. In this letter, we propose a novel method inspired by Majorization-Minimization (MM) to achieve efficient GMF fitting in a linear manner. The proposed method integrates the contribution of each Gaussian component in GMF to construct a linear surrogate and ensures the consistent convergence of the original nonlinear objective. Extensive experiments demonstrate that the proposed method outperforms classical solutions in convergence speed while maintaining precise fitting accuracy.

Index Terms—Gaussian mixture function, nonlinear regression, majorization-minimization, expectation-maximization, linear least squares.

I. INTRODUCTION

A linear superposition of Gaussian functions [1], the Gaussian Mixture Function (GMF) has been widely utilized in statistical signal processing and analysis, e.g., characterizing arterial blood volume pulse signals [2], [3], modeling stellar surface brightness [4], [5], and approximating the Probability Density Function (PDF) [6], [7]. Besides these, the GMF is also renowned for modeling non-Gaussian noise induced by intricate physical factors [8], [9]. Accordingly, fitting experimental data with the GMF to reasonably interpret the corresponding process

or phenomenon is a pivotal issue within the signal processing domain.

Since the GMF is essentially underlain by multiple exponential functions, the fitting of GMF to noisy observations is typically formulated as a nonlinear regression problem, which is cumbersome from both temporal and computational perspectives [10], [11], [12], [13], [14]. A general solution to this problem is nonlinear least squares algorithms [15], [16], [17], [18], [19]. These approaches usually employ derivatives of the regression objective to iteratively facilitate convergence. Specifically, the classical Newton-Raphson algorithm [19] utilizes both the gradient and Hessian matrix to numerically approximate the GMF parameters upon a suitable initial configuration. However, the second-order derivatives are expensive to calculate, which motivates the BFGS algorithm [15] to utilize gradients to approximate the Hessian matrix. Nevertheless, all these approaches are developed for generalized nonlinear regression problems and fail to estimate the GMF parameters in a closed-form solution through the involvement of function essentials, which may result in an inefficient fitting process.

In fact, not all GMF fitting methods follow this principle. In a special scenario of GMF utilized in PDF fitting, i.e., the Gaussian Mixture Model (GMM), a latent variable has been involved to change the derivatives-based nonlinear solution into a closed-form linear solution, which significantly increases the fitting accuracy and robustness [20]. In detail, the objective is reformulated as maximizing the log-likelihood in the famous Expectation-Maximization (EM) algorithm. A latent variable that denotes the posterior probability of each Gaussian component for the noisy observations is then evaluated using Bayes' theorem. Therefore, the log-likelihood can be maximized straightforwardly in closed form, which leads to an efficient convergence [21], [22], [23], [24]. Nonetheless, the EM algorithm cannot be directly extended to fit a non-normalized GMF, as it not only assumes a valid probability distribution, but also relies on the maximum likelihood estimation instead of regression-based fitting.

However, it is worth noting that the EM algorithm is a special case of the classical Majorization-Minimization (MM) algorithm [25], [26], [27]. In tackling the complicated GMF fitting problem, the MM algorithm can take advantage of the GMF property to provide a more efficient solution than a general-purpose solver. To exploit this advantage, this letter proposes a novel GMF Fitting (GMFF) method inspired by the iterative MM framework, named GMFF-MM. In detail, a latent variable denoting the contribution of each Gaussian component is introduced, which reformulates the original nonlinear objective as a linear surrogate objective. Then, a hyperparameter regulating the updated parameters is applied to the linear surrogate minimization,

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which guarantees consistent convergence of the original objective. Extensive experiments have demonstrated the feasibility and efficiency of the GMFF-MM. In conclusion, the proposed GMFF-MM is a linear solution that facilitates the closed-form estimation of GMF parameters from noisy observations with improved efficiency.

The rest of this letter is organized as follows: Section II introduces the theoretical derivation of the GMFF-MM. Section III presents experimental results to validate its efficiency. Section IV concludes this letter and outlines future work. Note that ‘ $\times$ ’ denotes the multiplication throughout this letter.

II. METHODOLOGY

A. Motivation

For easy discussion, a simple one-dimensional scenario is utilized in this paper. The noisy observations are defined as $\mathcal{D} = \{(x_j, y_j)\}_{j=1}^J$, in which y_j denotes the observed value at x_j . A Gaussian function can be written as:

$$\Phi(\theta; x) = A \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad (1)$$

where $x \in \mathbb{R}$ is the random variable, and $\theta = [A, \mu, \sigma]$ denotes the amplitude, mean, and standard deviation, respectively. Due to the underlying exponential function, directly estimating θ from noisy observations is cumbersome, as it requires solving a nonlinear regression problem. An effective way to counteract the exponential function is to take the natural logarithm of (1), resulting in the following polynomial [10]:

$$\begin{aligned} \log[\Phi(\theta; x)] &= \left(\log(A) - \frac{\mu^2}{2\sigma^2}\right) + \frac{\mu}{\sigma^2}x - \frac{1}{2\sigma^2}x^2 \\ &= a + bx + cx^2, \end{aligned} \quad (2)$$

where the unknown θ are transformed into the coefficients a , b and c in a linear form. Accordingly, by defining $y_j^* = \log(y_j)$, these coefficients can be explicitly derived using classical linear fitting methods, such as least squares [28]:

$$\arg \min_{a,b,c} \sum_{j=1}^J [y_j^* - (a + bx_j + cx_j^2)]^2. \quad (3)$$

Obviously, the closed-form solution of (3) exists, and θ can be simply determined through an inverse transform of (2). Note that (3) only considers observed values above zero, and a negative c is required to ensure a valid positive σ . However, the situation will be essentially different in GMF fitting. A GMF with N Gaussian components is defined as:

$$f(\Theta; x) = \sum_{n=1}^N \Phi(\theta_n; x) = \sum_{n=1}^N A_n \exp\left(-\frac{(x - \mu_n)^2}{2\sigma_n^2}\right), \quad (4)$$

where $\Theta = [\theta_1, \dots, \theta_N]$ denotes a parameter vector of multiple Gaussian components. Clearly, the similar simplification on single Gaussian cannot be directly applied to the GMF fitting problem. Taking the natural logarithm of (4) yields:

$$\log[f(\Theta; x)] = \log\left[\sum_{n=1}^N A_n \exp\left(-\frac{(x - \mu_n)^2}{2\sigma_n^2}\right)\right], \quad (5)$$

where the logarithm is unable to function on each Gaussian component since the summation effect. Then, the nonlinear

exponential nature remains inherently, requiring the GMF fitting to be formulated as a nonlinear regression problem:

$$R(\Theta) = \sum_{j=1}^J [y_j - f(\Theta; x_j)]^2. \quad (6)$$

Due to the complex form, the minimization of (6) has to be solved iteratively using a general Newton-type algorithm. In this case, the update rule for Θ is given by the computationally expensive gradient $\nabla R(\Theta^{(k)})$ and Hessian $H(\Theta^{(k)})$:

$$\Theta^{(k+1)} = \Theta^{(k)} - H^{-1}(\Theta^{(k)}) \nabla R(\Theta^{(k)}), \quad (7)$$

where k is the iteration index. With a suitable initial estimate, it is possible to obtain an optimal Θ^* that accurately fits the GMF to the noisy observations. Although there are several simplified variations [15], [16], [17], [18], a trade-off arises in requiring additional iterations for convergence. Hinted by GMM-based distribution fitting, this work aims to construct an efficient linear solution by involving a surrogate function.

B. Estimation With Majorization-Minimization

1) *Majorization Step:* With this aim, taking the natural logarithm of each term in (6) yields [10], [11]:

$$R^*(\Theta) = \sum_{j=1}^J [y_j^* - f^*(\Theta; x_j)]^2, \quad (8)$$

where $f^*(\Theta; x_j)$ is equal to (5). Inspired by the famous EM algorithm, a latent variable representing the contribution of each Gaussian component to the fitted GMF instead of the posterior probability can be defined as:

$$\omega_n(\Theta) = \frac{\Phi(\theta_n; x)}{\sum_{n=1}^N \Phi(\theta_n; x)}. \quad (9)$$

It is straightforward to conclude that $\sum_{n=1}^N \omega_n(\Theta) = 1$. Then, by simply substituting (9) into $f^*(\Theta; x)$, Jensen's inequality can be applied to derive a lower bound:

$$\begin{aligned} f^*(\Theta; x) &= \log\left[\sum_{n=1}^N \omega_n \frac{\Phi(\theta_n; x)}{\omega_n}\right] \\ &\geq \sum_{n=1}^N \omega_n \log\left[\frac{\Phi(\theta_n; x)}{\omega_n}\right] = g(\Theta|\hat{\Theta}; x), \end{aligned} \quad (10)$$

where $\hat{\omega}_n$ denotes the contribution related to $\hat{\Theta}$ for clarity, and $f^*(\Theta; x) = g(\Theta|\hat{\Theta}; x)$ holds if and only if $\Theta = \hat{\Theta}$.

With such modification, the logarithm is able to function on each Gaussian component rather than on the entire GMF. Therefore, $g(\Theta|\hat{\Theta}; x)$ can be simplified as the following polynomial after some basic rearrangements:

$$g(\Theta|\hat{\Theta}; x) = \sum_{n=1}^N \hat{\omega}_n (a_n + b_n x + c_n x^2) - \sum_{n=1}^N \hat{\omega}_n \log \hat{\omega}_n, \quad (11)$$

where the coefficients a_n , b_n , and c_n are now related to the parameters of each Gaussian component, following a similar mathematical formulation as (2). Consequently, the GMF parameters can be linearly solved by estimating these coefficients, leading to a tractable surrogate objective:

$$G(\Theta|\hat{\Theta}) = \sum_{j=1}^J [y_j^* - g(\Theta|\hat{\Theta}; x_j)]^2. \quad (12)$$

Suppose $\hat{\Theta}$ is known from the last k -th iteration, a lower bound $g(\Theta^{(k+1)}|\Theta^{(k)}; x)$ that fits the logarithm of the noisy observations can be derived by minimizing (12). Nevertheless, the naively estimated $\Theta^{(k+1)}$ is not enough to guarantee consistent convergence of the original objective (8), necessitating a revision to make (12) serve as a valid surrogate.

2) *Minimization Step*: As discussed, a hyperparameter $\beta^{(k)}$ is introduced over iterations to function on the noisy observations and a revised $\Theta^{(k+1)}$ can be obtained using:

$$\Theta^{(k+1)} = \arg \min_{\Theta} \sum_{j=1}^J \left[\beta^{(k)} y_j^* - g(\Theta|\Theta^{(k)}; x_j) \right]^2. \quad (13)$$

Essentially, the involvement of $\beta^{(k)}$ aims to guarantee that the original objective (8) is non-increasing in parallel with (12), which is achieved from the following inequality:

$$G(\Theta^{(k+1)}|\Theta^{(k)}) \geq R^*(\Theta^{(k+1)}). \quad (14)$$

By expanding and merging similar terms, (14) is rewritten as:

$$\begin{aligned} & \sum_{j=1}^J \left[g(\Theta^{(k+1)}|\Theta^{(k)}; x_j) - f^*(\Theta^{(k+1)}; x_j) \right] \\ & \times \left[g(\Theta^{(k+1)}|\Theta^{(k)}; x_j) + f^*(\Theta^{(k+1)}; x_j) - 2y_j^* \right] \geq 0. \end{aligned} \quad (15)$$

Let $\tau^{(k)} = f^*(\Theta^{(k+1)}; x) - g(\Theta^{(k+1)}|\Theta^{(k)}; x)$, and $\eta^{(k)}$ denotes the estimation error from solving (13). A simplified expression can be obtained as follows:

$$(1 - \beta^{(k)}) \sum_{j=1}^J y_j^* \tau_j^{(k)} \geq \frac{1}{2} \sum_{j=1}^J \tau_j^{(k)} (2\eta_j^{(k)} + \tau_j^{(k)}). \quad (16)$$

Since $\tau^{(k)} \geq 0$ always holds by Jensen's inequality, an upper bound constraint on $\beta^{(k)}$ when $\forall y_j \geq 1$ is derived as:

$$\beta^{(k)} \leq 1 - \frac{\sum_{j=1}^J \tau_j^{(k)} (2\eta_j^{(k)} + \tau_j^{(k)})}{2 \sum_{j=1}^J y_j^* \tau_j^{(k)}}, \quad \forall y_j \geq 1. \quad (17)$$

In contrast, (17) needs to be reversed to a lower bound constraint when $\forall y_j < 1$. As a result, the constraint on $\beta^{(k)}$ is specified according to the values of the observations, and the non-increasing property of (8) can be ensured by designing an appropriate $\beta^{(k)}$ to satisfy this constraint. It has been revealed that (12) is non-increasing over iterations by iteratively regulating $\beta^{(k)}$ towards 1. Then, linear least squares can be utilized to readily solve (13) by expanding it over the observations:

$$\begin{aligned} & \begin{bmatrix} \omega_{1,1}^{(k)} & \omega_{1,1}^{(k)} x_1 & \dots & \omega_{N,j}^{(k)} x_j^2 \\ \vdots & \vdots & \ddots & \vdots \\ \omega_{1,J}^{(k)} & \omega_{1,J}^{(k)} x_J & \dots & \omega_{N,J}^{(k)} x_J^2 \end{bmatrix} \begin{bmatrix} a_1^{(k+1)} \\ b_1^{(k+1)} \\ \vdots \\ c_N^{(k+1)} \end{bmatrix} \\ & = \begin{bmatrix} \beta^{(k)} y_1^* + \sum_{n=1}^N \omega_{n,1}^{(k)} \log \omega_{n,1}^{(k)} \\ \vdots \\ \beta^{(k)} y_J^* + \sum_{n=1}^N \omega_{n,J}^{(k)} \log \omega_{n,J}^{(k)} \end{bmatrix}. \end{aligned} \quad (18)$$

where J is assumed to be greater than the number of coefficients $3N$. After solving (18) for coefficients $a_n^{(k+1)}$, $b_n^{(k+1)}$ and $c_n^{(k+1)}$,

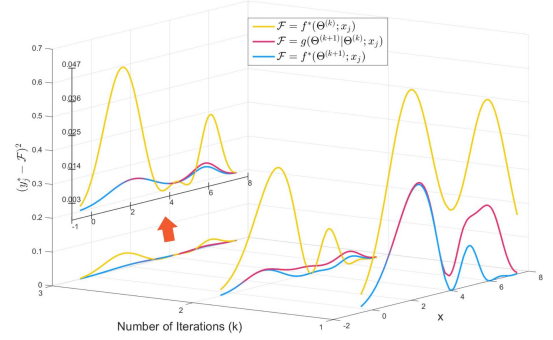


Fig. 1. The evolving behavior of the surrogate and original objectives.

the desired parameters $\theta_n^{(k+1)}$ for each Gaussian component are simply determined as:

$$\begin{aligned} \mu_n^{(k+1)} &= \frac{-b_n^{(k+1)}}{2c_n^{(k+1)}}, \\ \sigma_n^{(k+1)} &= \sqrt{-\frac{1}{2c_n^{(k+1)}}}, \\ A_n^{(k+1)} &= e^{a_n^{(k+1)} - \frac{(b_n^{(k+1)})^2}{4c_n^{(k+1)}}}. \end{aligned} \quad (19)$$

Now, the GMFF-MM is ready for use, with the initial estimate derived from the K-means algorithm. The surrogate (12) is constructed by updating the contribution terms based on (9) in the Majorization step, and the desired parameters are computed according to (19) with an appropriate $\beta^{(k)}$ in the Minimization step. Note that both $\tau^{(k)}$ and $\eta^{(k)}$ are *a posteriori* quantities, and the addition of $\beta^{(k)}$ may not consistently satisfy its constraints. Nevertheless, the violation of (14) provides a clear indication of convergence failure, which enables us to introduce a simple protection. Specifically, a subtle random disturbance ζ is imposed on the observations, upon which the K-means algorithm is applied to generate a modified initial estimate. This modified estimate serves as a new initialization for the GMFF-MM algorithm, which then iteratively fits the original observations until convergence. The final criterion of convergence can be determined by the Mean Squared Error (MSE) $R(\Theta)/J$, where an extremely small threshold ϵ is employed to numerically assess whether convergence has been approached.

III. EXPERIMENT RESULTS AND DISCUSSION

In this section, the performance of the proposed GMFF-MM has been validated through a series of representative experiments. Fig. 1 illustrates the iterative evolution of the surrogate and original objectives, visually demonstrating that an appropriate $\beta^{(k)}$ can ensure consistent convergence. The red curve indicates the surrogate objective evaluated at each observed point, while the blue curve represents the original $[y_j^* - f^*(\theta^{(k+1)}; x_j)]^2$, and the yellow curve represents $[y_j^* - f^*(\theta^{(k)}; x_j)]^2$ from the previous k -th iteration. As observed, the red curve invariably lies above the blue curve, which demonstrates that (14) remains valid throughout the iterations. Furthermore, both the surrogate and original objectives exhibit consistent convergence as $\beta^{(k)}$ is iteratively regulated towards 1, indicating that properly designing $\beta^{(k)}$ to satisfy the bound

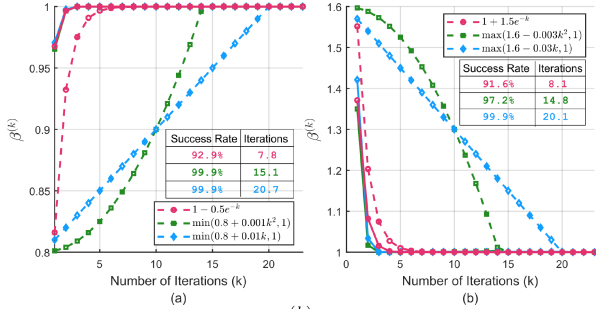


Fig. 2. The empirically designed $\beta^{(k)}$ compared with its bounds: (a) upper bound when $\forall y_j \geq 1$; (b) lower bound when $\forall y_j < 1$. Two metrics in the tables are evaluated by 10^4 simulation trials: (i) success rate denotes the percentage of trials that converged without protection; (ii) iterations denote the average number of iterations required for convergence with protection.

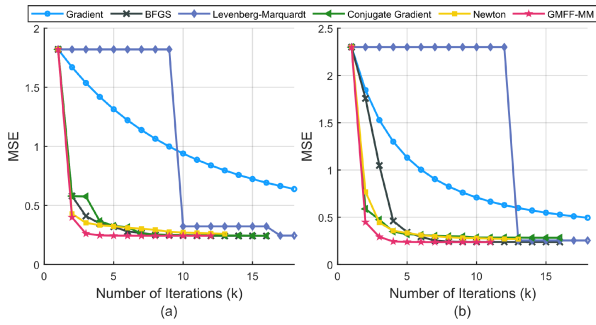


Fig. 3. The MSEs of GMFF-MM and nonlinear optimization algorithms in fitting GMFs with (a) two and (b) three Gaussian components.

constraint ensures the effectiveness of GMFF-MM. Fig. 2 illustrates a comparison between the empirically designed $\beta^{(k)}$ and their bound constraints, including exponential, quadratic and linear. In Fig. 2(a), the coefficients $\beta^{(k)}$ are designed to fit a GMF by considering observed values above 1, while Fig. 2(b) shows the case where observed values below 1. The dashed lines correspond to the three empirical configurations, and the solid lines denote the upper or lower bound constraints for each configuration. It can be noted that these empirical configurations are bounded by their respective constraints, which indicates that a well-fitted GMF can be expected. However, an optimized $\beta^{(k)}$ depends on *a posteriori* quantities, and GMFF-MM has incorporated a protection to ensure robustness. Accordingly, 10^4 simulation trials have been conducted to obtain the quantitative results summarized in the embedded table. The success rate denotes the percentage of trials that converged without protection and the iterations denote the average number of iterations required for convergence with protection. The convergence threshold ϵ is set as $1e-7$. The success rate ranges from 91.6% to 99.9% across different configurations, demonstrating the robustness of GMFF-MM in GMF fitting. Notably, the exponential $\beta^{(k)}$ achieves faster convergence than the other two configurations, benefiting from its closest alignment with the bound constraint. Therefore, $\beta^{(k)} = 1 - 0.5e^{-k}$ is employed to further validate the performance of GMFF-MM.

In Fig. 3, a comparative analysis is presented between GMFF-MM and several classical nonlinear optimization algorithms [15], [16], [17], [18], [19] in fitting the GMF to the noisy

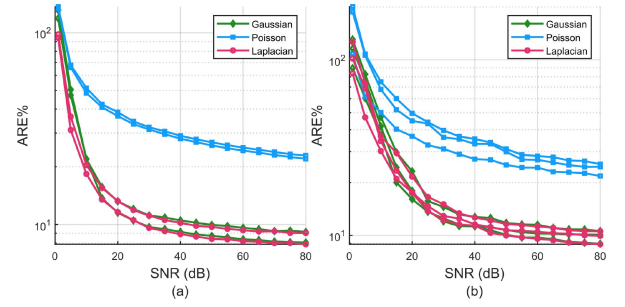


Fig. 4. The AREs of estimated parameters versus the SNR in fitting GMFs with (a) two and (b) three Gaussian components.

observations. The observed values are generated by contaminating the theoretical values with zero-mean Gaussian noise of standard deviation $\sigma_n = 0.5$. Fig. 3(a) illustrates the MSE results for fitting a GMF with two Gaussian components, and Fig. 3(b) corresponds to the case with three components. It is evident that the proposed GMFF-MM outperforms the classical solutions in terms of convergence speed across both fitting trials. This advantage arises from GMFF-MM explicitly accounting for the contribution of each Gaussian component, which enables a more efficient linear solution compared to general-purpose solvers. In addition, GMFF-MM exhibits fitting accuracy comparable to the other algorithms, with converged MSEs of approximately 0.24 and 0.27, respectively. In Fig. 4, the Absolute Relative Error (ARE) across the Signal-to-Noise Ratio (SNR) domain is illustrated to evaluate the robustness of the proposed GMFF-MM. The ARE percentile [12] for each Gaussian component is given as $\|\theta_n^* - \theta_n^+\|_1 / \|\theta_n^+\|_1 \times 100\%$, where θ^* and θ^+ represent the estimated and true parameters, respectively. Here, $|\cdot|$ denotes the absolute value and $\|\cdot\|_1$ denotes the ℓ_1 -norm. The SNR ranges from 1 to 80, with 10^4 simulation trials performed at each noise level to obtain statistical ARE results. Clearly, the AREs decrease with increasing SNR in both fitting cases, where Fig. 4(a) shows the results for fitting two Gaussian components, and Fig. 4(b) corresponds to fitting three Gaussian components. Moreover, three noise models are employed to provide a comprehensive evaluation, including the Gaussian, Poisson, and Laplacian models. As expected, Poisson noise results in inferior performance compared to the other two noise models. This is because GMFF-MM is essentially a linear least squares method that assumes a symmetric noise distribution. Nevertheless, GMFF-MM demonstrates robust performance under both Gaussian and Laplacian noise, as evidenced by a rapid decrease in AREs as the SNR increases to 40 dB, followed by stabilization at higher SNR levels.

IV. CONCLUSION

In this letter, a novel method named GMFF-MM is proposed to solve the GMF fitting problem in a linear manner, where a latent variable is defined to construct a simplified surrogate, and a hyperparameter β is designed to ensure consistent convergence. Experimental results validate the efficiency even without an optimized β . Further improvements in accuracy and efficiency are expected with an analytically optimized β , which will be discussed in our future work. Accordingly, GMFF-MM serves as a valuable signal processing technique that efficiently fits experimental data with GMF to facilitate reasonable interpretation of the corresponding process.

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